VIRAL INFECTIONS

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The main viral diseases of the skin are:

- Herpes simplex.
- Varicella zoster.
- Warts.

Herpes virus group

- **DNA viruses**, replicate **intranuclear**.
- **Virus persists throughout life** as a latent infection in one of a sensory ganglia.
- Under **immune suppression**, the virus replicates and produce infection.

Herpes simplex

Etiology

- caused by ***Herpesvirus hominis*** **(HSV).**

- The spread of infection:

Direct contact with person with an active infection or droplets from infected secretions

- IP: **2-12 days.**

Clinical picture

1 – Primary herpes simplex

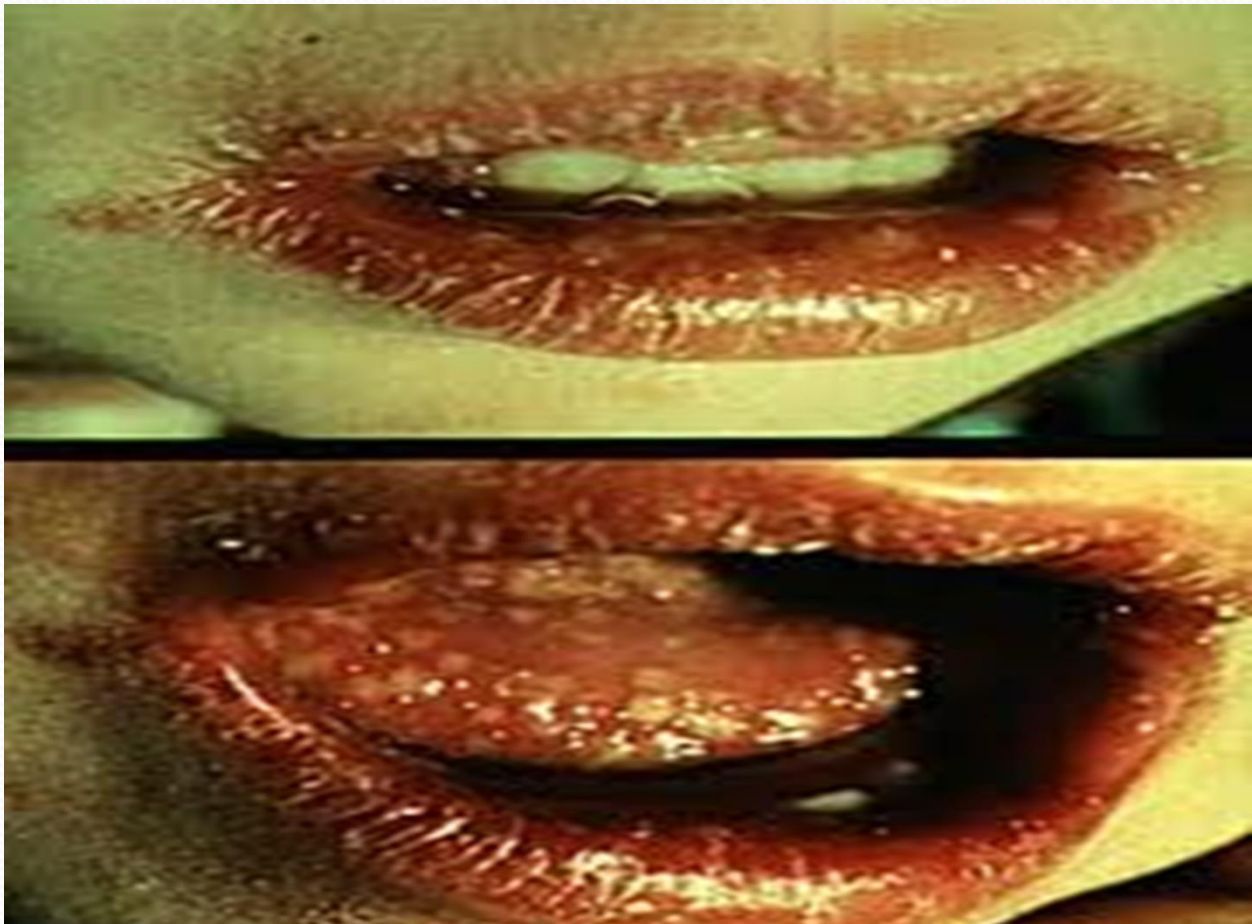
- Primary infection (for the first time).
- usually subclinical, but when clinical lesions develop, the severity is generally greater than in recurrences.
- occurs mainly in infants & young children

Clinical types

- Herpetic gingivostomatitis
- Herpes genitalis
- Keratoconjuctivitis
- Neonatal herpes simplex

Clinical types

- **Herpetic gingivostomatitis**: painful vesicles & ulcers



- **Herpes genitalis**
  - It occurs after the onset of sexual activity

- **Keratoconjunctivitis**





- **Neonatal herpes simplex:** acquired through an infected birth canal of the mother.



2- Recurrent herpes simplex

- The virus may be **reactivated** in **previously infected** person

Triggering factors

* Trauma
* Ultraviolet radiation
* Surgery
* Emotional stress
* Menstruation
* Infections e.g., upper respiratory tract infections.

Clinical picture

Sudden onset of burning sensation precedes appearance of a swollen red patch with small grouped vesicles.

After 1-2 days → vesicles dry → crusts → healing in 7-10 days **without scarring.**





Clinical types

- Herpes facialis (labialis)
- Herpetic keratoconjunctivitis
- Herpes genitalis







Herpes facialis (labialis)



Herpetic keratoconjunctivitis



Herpes genitalis

Treatment

A. Topical treatment

- **Drying lotions** for the vesicles.
- **Antibiotics** for 2ry infection.
- **Acyclovir** cream.

B. Systemic treatment

- **Acyclovir** is the treatment of choice.
- Newer antivirals e.g. valacyclovir and famciclovir.

Varicella-zoster infection

- 1ry infection → varicella (Chickenpox)
- Reactivation of a latent virus → Herpes zoster

1-Varicella (Chickenpox)

Incubation period is 10-21 days

Usually affects children

Transmission is by → direct contact

→ droplet infection (respiratory route)

Infection gives **permanent immunity**

Clinical picture

- low grade fever, headache & malaise

- *Successive crops* macules then vesicles then pustules then crusts are formed → pleomorphic eruption (lesions in different stages)

Site: trunk, face & oral mucosa(centripetal).

- Mild Pruritus.

Treatment

- Antipruritic lotions & **antihistamines** for itching.
- **Antibiotics** for 2ry bacterial infection.
- **Systemic antiviral** drugs in severe cases.





2-Herpes zoster

- A reactivation of the latent virus.

- The **middle-aged and elderly** individuals are most often affected.
- more frequent in *immunocompromised* persons.

Clinical picture

- **Pain** is the first manifestation of the disease, may be severe.
- After 2-3 days → **groups of vesicles** on an erythematous base **within the distribution of a sensory nerve.**
- The vesicles become **pustular, crust** and heal in 2-4 weeks.
- The eruption is usually **unilateral** and **affects one dermatome**



Lumbosacral HZ





Thoracic HZ

Ophthalmic zoster (Herpes zoster ophthalmicus)



Complications

- **Post-herpetic neuralgia**. persistence of pain more than one month after the onset of the disease
- **Ocular complications.**
- **Secondary bacterial infection.**
- **Facial palsy.**

Treatment

Topical treatment

- As for herpes simplex.

Systemic treatment

- Analgesics for pain.
- Antibiotics for secondary bacterial infection.
- **Acyclovir** and the newer antivirals e.g. valacyclovir and famciclovir

Warts (Verrucae)

-Common infectious skin growths caused by ***Human papilloma virus*** **(HPV).**

-occur more in children & young adults .

Mode of transmission

- Warts spread by direct or indirect contact.
- Trauma and maceration predispose to inoculation of the virus.

Clinical types

1- Common warts

2- Plane warts

3- Filiform warts

4- Digitiform warts

5- Plantar warts

6- Anogenital warts (sexually transmitted)

Common warts:

- **papules with rough, horny surface**
- most commonly on the **back of the hands and fingers.**
- usually *symptomless*.



Plane warts (flat warts):

- **Multiple, smooth, flat** lesions usually skin-colored.
- The **face and back of the hands** are the sites of predilection.



Filiform warts:



Digitiform warts:



Plantar warts:

- affect the **sole of the foot**.
- **painful** well defined rounded skin lesion with rough surface.



Treatment

1- Chemical treatment:

- Salicylic acid 15-20%
- Podophyllin 10-25% or imiquimod cream for anogenital warts

2- Electrocautery.

3- Cryotherapy with liquid nitrogen

4- Laser therapy.

5-Immunotherapy (MMR vaccine, candida vaccine).


Thank
You